
EM 653: Energy Harvesting

Course Code	EM 653
Course Name	Energy Harvesting
L-T-P-C	3-0-0-3
Intended for	M.Tech Energy Engineering
Distribution	Specialisation elective
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Course Modules

- **Module 1 (Thermoelectric Materials):** General Principles and Theoretical Considerations: Thermoelectric Phenomena, Conversion Efficiency and Figure-of-Merit, Thermoelectric Transport Theory, Optimization of Carrier Concentration, Minimizing the Thermal Conductivity, Selective Carrier Scattering in Thermoelectric Materials Material Preparation and Measurement of Their Thermoelectric Properties, Z-Meters, Methodology for Testing Thermoelectric Materials and Devices Thermoelectric Materials, Applications and Devices (15)
- **Module 2 (Piezoelectric Materials):** Piezoelectric materials, Structures of piezoceramics and different groups of piezoceramic materials. General understanding of piezoelectric properties, principles and their influence, Processing of piezoceramic materials (bulk, film and composite), Different processing techniques, Characterization of physical properties, Piezoceramics bulk applications and piezocomposite applications - advantages and disadvantages, Electrical characterization of piezoelectric materials, poling techniques, electroding, measurements of piezoelectric properties of bulk ceramics/ piezocomposites. (15)
- **Module 3 (Application of Piezoelectric Materials):** Various kinds of actuators, sensors, piezoelectric and electromagnetic Energy Harvesting.- Piezoelectric Energy Harvesting.- Electromechanical Modeling of Cantilevered Piezoelectric Energy Harvesters for Persistent Base Motions.- Performance Evaluation of Vibration-Based Piezoelectric Energy Scavengers.- Piezoelectric Equivalent Circuit Models.- Electromagnetic Energy Harvesting.- Energy Harvesting Circuits and Architectures.- On the Optimal Energy Harvesting from a Vibration Source Using a Piezoelectric Stack.- Energy Harvesting Wireless Sensors.- Energy Harvesting using Non-linear Techniques.- Power Sources for Wireless Sensor Networks.- Harvesting Microelectronic Circuits. Stimulated Muscle-Powered Piezoelectric Generator as a Power Source for Implanted Medical Devices.- Piezoelectric Energy Harvesting for Bio MEMS Applications.- Harvesting Energy from the Straps of a Backpack Using Piezoelectric Materials.- Energy Harvesting for Active RF Sensors and ID Tags.- Powering Wireless SHM Sensor Nodes through Energy Harvesting. (12)

Text Books

1. D. M. Rowe, *CRC Handbook of Thermoelectrics*, 1995.
2. Shashank Priya and Daniel J. Inman, *Energy Harvesting Technologies*, Springer, 2009.